



DET NORSKE VERITAS

DESIGN VERIFICATION REPORT

Designer/ Manufacturer : SPECIAL AND SAFETY WORK (SSW),
Riyadh, Saudi Arabia

DNV Project No : PP085728

Equipment Description : Anti Blast Resistant Door, Double Leaf
SSW Drawing no: DN-01 (Sht 1 of 1) Rev-01

Design Codes and Standards : AISC 9th Edition (as applicable)

This Is To Certify That:

The structural design of the Double Leaf, Anti Blast Resistant Door (SSW Drawing no: DN-01 (Sht 1 of 1) Rev-01) has been reviewed and design complies with the Design Code AISC 9th Edition (as applicable) with regards to local in-place strength against uniformly distributed static equivalent peak over pressure of 3psig.

Validity Period:

No validity period is stated on the design verification report as it only confirms to the conformity with specified regulations/ standards at the time of issue. Change in regulations or change in design may have influence on validity.

Scope of Verification:

Verification of the design is limited to review of structural analysis submitted for compliance with AISC (9th Edition) as applicable. No other codes or standards are referred to during this review. Client to satisfy themselves regarding the requirements of other relevant codes / standards / national or local regulations if applicable.

Below mentioned parameters are considered for design review –

Peak Over Pressure	3 psig
Reflected Pressure	5 psig
Rebound Pressure	50% of reflected pressure
Type of Door	Hinged type, with Double Leaf
Overall Frame Size	2150 mm (SOW) x 3075 mm (SOH)

If any person suffers loss or damage which is proved to have been caused by any negligent act or omission of Det Norske Veritas, then Det Norske Veritas shall pay compensation to such person for his proved direct loss or damage. However, the compensation shall not exceed an amount equal to ten times the fee charged for the service in question, provided that the maximum compensation shall never exceed USD 300,000. In this provision "Det Norske Veritas" shall mean the Foundation Det Norske Veritas as well as all its subsidiaries, directors, officers, employees, agents and any other acting on behalf of Det Norske Veritas.



This review does not cover the design and integrity of concrete structure around the door. The suitability of the door components against fire resistance, their quality and suitability is not considered part of the present scope. The in-service deterioration of the door components due to various causes like vibration, thermal expansion, contraction, wear, etc. is not examined by DNV as part of this report. Protection from blast fragment and impact loading are not considered as part of the present scope of verification. Fabrication and method of installation is also excluded from the scope of this report.

Conditions:

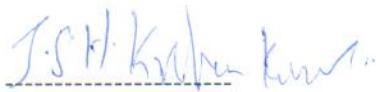
1. The design appraisal is limited to the static in place analysis of the steel door frame and excludes any aspects towards adequacy of anchoring arrangement. The scope of verification is listed as above.
2. Door face steel sheet must have yield strength not less than 250 MPa (36259 psig) as used in design calculation. All material used for door and the hardware components like door frame anchors, tower bolts, hinges, etc. including sealants and panic devices shall meet the blast rating specifications and structural quality as per the client requirements.
3. Present appraisal has been carried out with the blast load indicated in the submitted calculations upon confirmation from the designer that no other loads/ load factors are applicable.
4. Manufacturer to ensure that fabrication, installation and adjustment of various hardware components of the door will be as per client specifications and should be able to meet performance requirements stipulated by the client.
5. Any aspects related to inspection and testing are excluded from the review.

List of Documents:

S. No.	Document Title	Document No.	Rev.	Status
1	Anti Blast Resistant Door	DN-01 (Sht 1 of 1)	Rev-1	Approved

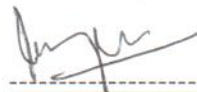
for DET NORSKE VERITAS AS, Abu Dhabi Branch

Name:



Krishna kumar J.S.H
Static Pressure Design Engineer





Rajeev Sharma
Service Lead - Design Appraisal and Analysis Services

Date: 5th November 2013
Place: Dubai , UAE